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Effects of photobiomodulation therapy on orthodontically induced inflammatory root resorption: a systematic review and meta-analysis

Guanning Zhang[1], Zhuoying Li[1], Colman McGrath[2], Yanqi Yang[1] and Zhiyi Shan[1*]

Abstract

Introduction Orthodontically induced inflammatory root resorption (OIIRR) is a common complication of orthodontic treatment that may compromise tooth health and treatment outcomes. Photobiomodulation (PBM) therapy has emerged as a potential modality to mitigate against OIIRR. However, clinical studies have shown inconsistent efficacy and there is a lack of consensus on functional parameters, underscoring the need for evidence-based reviews.

Methods Randomized controlled trials (RCTs) investigating PBM for OIIRR mitigation with quantitative root resorption assessment were included in the review. An electronic search was conducted across six databases (PubMed, Web of Science, Cochrane Library, Embase, MEDLINE, and Scopus) and the OPENGREY.EU for grey literature. References of included articles and relevant reviews were manually screened. The search was last updated on May 27th, 2025, with no publication date restrictions. The quality of included studies and evidence certainty of outcomes were assessed using the Cochrane Risk of Bias Tool 2.0 (RoB 2) and the 'Grading of Recommendations, Assessment, Development and Evaluations' (GRADE) framework. Following the heterogeneity evaluation, meta-analysis was performed on eligible studies using RevMan 5.3 software. Inter-rater agreement during study selection and quality assessment was quantified using Cohen's kappa statistics. The significance threshold was set at $p < 0.05$.

Results From a total of 893 records, 13 RCTs were included in the systematic review, with 4 selected for meta-analysis. Among all the included studies, only two demonstrated low risk of bias. Possible functional settings of PBM for OIIRR varied across studies (output power 100–200 mW, energy density 5–10 J/cm^2, and per-session dose 10–100 J/tooth). Within these ranges, appropriate cumulative doses and a tapered irradiation frequency protocol could influence PBM efficacy. The meta-analysis revealed that PBM was associated with a statistically significant ($p < 0.01$) but modest reduction (MD: -0.13 mm^3, 95% CI: -0.17 to -0.09 mm^3) in OIIRR of maxillary first premolars undergoing buccal tipping movement.

*Correspondence:
Zhiyi Shan
shanzhiy@hku.hk

Full list of author information is available at the end of the article

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Conclusions PBM shows potential in reducing OIIRR in the maxillary first premolar during buccal tipping, especially in the early stages of treatment. However, variability in PBM parameters and limited studies with low risk of bias indicate the need for standardized protocols and higher-quality research.

Keywords Photobiomodulation, Orthodontic tooth movement, OIIRR

Introduction

Orthodontically induced inflammatory root resorption (OIIRR) is an inevitable and unpredictable pathological side effect that can occur during orthodontic treatment [1, 2]. The incidence of OIIRR can be high [3]. Some degree of root resorption is considered normal [4], while excessive resorption, which occurs in 1% to 5% of the teeth after orthodontic treatment, can compromise tooth stability and long-term health [5, 6]. OIIRR typically begins at the cellular level within 1–3 weeks of force application [7]. Early lesions can be identified within 4–8 weeks using high-resolution methods such as micro-CT or SEM, and macroscopic apical shortening becomes more reliably visible on conventional periapical radiographs only after several months [6]. Studies have found many risk factors that correlated with the severity of OIIRR, such as duration of orthodontic treatment, orthodontic force, and distance of tooth movement [8, 9]. To reduce OIIRR risk, treatment strategies emphasize controlled mechanics, shorter duration, and individual factors like genetics, age, and trauma history [10]. Other interventions such as corticotomy and piezocision used to aid orthodontic tooth movement [11, 12], may avoid the potential risk of OIIRR, but can lead to higher cost, pain, and have possible side effects of surgery [13].

In recent years, photobiomodulation (PBM) therapy, also known as low-level laser therapy (LLLT), has been gradually used in orthodontic clinical operations because of its unique advantages [14]. PBM therapy is a non-invasive light therapy using low-intensity lasers or light-emitting diodes (LEDs) in the visible and near-infrared spectrum with wavelength ranging from 620 nm to 1440 nm [15]. By acting on mitochondria, PBM increases adenosine triphosphate (ATP) production, modulates reactive oxygen species (ROS), and influences growth factor signaling [16], leading to transcriptional responses that support tissue repair, reduce inflammation, and promote cell proliferation and migration [15, 17, 18].

Several studies have examined the effects of PBM on OIIRR. In animal research, a commonly used model is inducing mesial movement of molars in rats and rabbits, with PBM used as an adjunctive treatment [19, 20]. The results indicated that PBM may reduce OIIRR, potentially through modulation of the OPG/RANKL pathway. Clinical studies, including randomized controlled trials (RCTs), have also been available in recent years [21, 22]. These studies incorporated diverse orthodontic tooth movement models, with variations in tooth type, applied force magnitude, movement direction, and treatment duration. Furthermore, PBM was delivered as adjunctive therapies with different parameters and protocols. Clinical trial outcomes are inconsistent. Some studies found substantial reductions in root resorption and others found no benefit. Prior systematic reviews reported the same pattern [13, 23].

PBM parameters like wavelength, output power, irradiance, fluence, exposure time, and irradiated area are critical determinants of treatment efficacy across various dental applications [24–26]. Previous reviews on orthodontic tooth movement (OTM) not only confirm that PBM efficacy varies with parameter choices but also highlight inconsistent reporting that disturbs the reproducibility [27–30]. By contrast, OIIRR-specific evaluation is much scarcer and carries additional limitations. The heterogeneity of PBM protocols and OIIRR outcome measures has prevented meaningful synthesis, leaving no consensus on efficacy or optimal settings of PBM on OIIRR. More robust evidence is needed to address this gap.

This systematic review evaluated PBM's effects on OIIRR through qualitative and quantitative synthesis of RCT results. Our aim was to deliver evidence-based recommendations for OIIRR management and to define clinically favorable parameter ranges by appraising PBM settings.

Methods

Protocol and registration

The systematic review was conducted and reported following the Cochrane handbook for systematic review and the checklist of Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) [31–33]. The protocol of the systematic review was registered in PROSPERO on 21 st January 2024 (CRD 42024500344).

Eligibility criteria

The eligibility criteria were determined by the PICOS (Participants, Intervention, Comparison, Outcomes, Study Design) strategy.

P (Participants): Patients of any age, sex, or ethnicity undergoing fixed orthodontic treatment for any type of tooth movement and treatment stage were included. Excluded patients with systemic disease or undertaking systemic medications, intraoral surgery, or orthognathic surgery that may affect bone metabolism. Studies conducted on replanted teeth, or teeth with previous

orthodontic treatment, current root resorption, trauma, root canal treatment, infected pulp, apical periodontitis, and periodontal diseases were not considered either.

I (Intervention): Using low-level laser or light-emitting diode for PBM therapy.

C (Comparison): Patients undergoing orthodontic treatment without PBM therapy.

O (Outcomes): Reporting quantitative measurement of OIIRR.

S (Study Design): Studies designed as randomized controlled trials were included. Both split-mouth and parallel designs were eligible. The minimum follow-up duration required 28 days (four weeks). Review articles, case reports, case series, letters, and studies only available in non-English versions were excluded.

Information sources and search strategy

The search procedure was carried out with no restrictions on the date published. Six electronic databases, PubMed, Web of Science, Cochrane Library, Embase via Ovid, MEDLINE via Ovid, and Scopus, were searched for eligible studies based on a combination of four keywords (orthodontics, root resorption, low-level laser therapy, dose) and their synonyms with Boolean operators. The search was also optimized using MeSH and Emtree terms. The detailed search strategy was listed in the supplementary (**Table S1-S6**). To avoid omissions, we included OPENGREY. EU to identify grey literature and examine reference lists of relevant reviews and included articles. The pooled results were exported to EndNote 21 software (Clarivate Analytics, Philadelphia, Pa) for duplicate removal and record screening. The search was initiated in March 2024 and updated in May 2025.

Study selection and data collection

Two review authors (G.Z. and Z.L.) screened the titles and abstracts independently, then assessed full texts for potentially eligible articles. Any disagreement was resolved by discussion with the third author (Z.S.). Studies that conformed to the eligibility criteria were collated for data extraction, including author names, publication year, study design, description of participants, orthodontic interventions, design of comparisons, and outcome measurements. Data related to PBM parameters was also extracted and listed based on light type, irradiation mode, wavelength, output power, power density, energy density, irradiation point, irradiation area, irradiation time, frequency, per-session dose per tooth, and treatment duration.

Outcomes and prioritization

The primary outcome of the systematic review was to evaluate the effectiveness of PBM therapy on OIIRR quantitatively and qualitatively. For quantitative synthesis, the review pooled data of OIIRR reporting as three-dimensional (3D) resorption crater volume (mm^3) from micro-computed tomography (micro-CT) images after assessing the calibration and reliability. Subgroup analysis was conducted by different axial root surfaces. For qualitative synthesis, the review narratively summarized PBM's effects on OIIRR stratified by type of tooth movement. Positive effects were defined as statistically significant differences in OIIRR based on 2D/3D measurements (mm, mm^3; quantification) or Levander and Malmgren index (grading) from periapical digital radiographs, CBCT, or micro-CT. The secondary outcome evaluated how PBM parameters (wavelength, output power, energy density, per-session dose, etc.) influence OIIRR during orthodontic tooth movement. The effects were discussed narratively because sufficient comparable data was not available.

Risk of bias in individual studies

The two review authors (G.Z. and Z.L.) evaluated the risk of bias (RoB) independently using the Cochrane Risk of Bias tool 2.0 (RoB 2) for included RCTs [31], and discuss the inconsistent opinions with the third author (Z.S.). Each study was categorized as low (indicating low risk of bias across all key domains), some concerns (indicating "some concerns" risk of bias in at least one key domain), or high (indicating high risk of bias in at least one key domain) risk of bias.

Data synthesis

A narrative synthesis of the extracted data was implemented regarding population characteristics, type of tooth movement, parameters and treatment protocols of PBM therapy, etc. for studies adopting both parallel and split-mouth design. Studies with the same outcome metrics under similar orthodontic interventions were involved in a quantitative synthesis based on methodological heterogeneity using RevMan 5.3 software. Because all trials included in the meta-analysis used a split-mouth design, outcomes were analyzed as paired data to account for within-patient correlation [34]. According that all outcomes were continuous, for each study we calculated the paired mean difference (MD) between experimental and control sites and its standard error (SE), and pooled effects using generic-inverse-variance methods with 95% confidence intervals (CI). When the standard deviation of the within-patient differences (SD_diff) was not reported, we derived it from the arm-specific SDs using an assumed within-person correlation ($R = 0.50$) informed by a similar study [35], and then computed the SE as SD_diff/sqrt(n) (n = the number of pairs) [34]. Sensitivity analysis was conducted using alternative R values (0.30 and 0.70) to evaluate the robustness of the results. Random-effects model was employed due

to potential heterogeneity in PBM treatment protocols. Heterogeneity between the studies was estimated using tau-squared (Tau^2), Cochran's Q test, and I^2 index with statistical significance indicated by $p < 0.05$. The 'Grading of Recommendations, Assessment, Development and Evaluations' (GRADE) tool was used by the two review authors (G.Z. and Z.L.) independently to evaluate and rank the outcome evidence certainty of meta-analysis and included records [36]. Cohen's kappa statistics were used for inter-rater agreement evaluation.

Results

Study selection

893 records were retrieved from all databases and imported into EndNote 21 software. After removal of duplicates, 451 records were screened for titles and abstracts according to the eligibility criteria. Some 434 records were then excluded for several reasons, including non-research articles (ongoing clinical trials, reviews, and case reports), non-clinical studies (in vitro and animal experiments only), and reports that did not measure root resorption as an outcome. Of the remaining 17 articles, four studies were excluded after retrieving and reviewing their full text: three with non-RCT experimental designs [21, 22, 37] and one lacking quantitative analysis of root resorption [38]. A substantial agreement was confirmed between the two review authors in the full-text-inclusion procedure ($\kappa = 0.850$). Manually screening of citations from included studies and relevant reviews didn't reveal any additional reports. The search for grey literature didn't yield any reports either. Ultimately, 13 articles were included in the systematic review. Among them, four articles that employed the same methods of orthodontic intervention and outcome measures were incorporated into quantitative analysis after heterogeneity assessment [35, 39–41]. One study [42] only mentioning "randomly divided" without adequate randomization methodological details was excluded from meta-analysis due to high risk of bias but included in qualitative synthesis. The literature selection process was shown in the PRISMA flowchart (Fig. 1).

Risk of bias

The agreement between the two reviewers using RoB 2 was excellent ($\kappa = 0.891$). The results of the RoB analysis are presented in Figs. 2 and 3. Among the 13 included records, only two were classified as having a low risk of bias [35, 39]. Five records were categorized as having "some concerns" due to the blindness issue from participants and operators [40, 41, 43–45]. The awareness of different interventions among patients and dentists might influence their behavior and attitudes, potentially introducing bias into the study outcomes. Six records were classified as having a "high risk" of bias mainly because the blindness issue from outcome assessors [42, 46–49]. Among these, one record didn't address the randomization methods and implementation, and reported notable differences in baseline characteristics between the experimental and control groups [49]. The other record was rated as "high risk" due to failure to specify statistical approaches or reported unvalidated results [50].

Characteristics of included studies

A summary of the basic information profile, participants, and experimental details of the 13 included studies is presented in Table 1. The publication years of these studies ranged from 2018 to 2024. The number of participants included in each study varied from 10 to 32, with ages ranging from 13 to 60 years. Most studies included both males and females with a varied sex ratio, and two studies exclusively included females [43, 45]. Except for two studies that experienced patient loss to follow-up [45, 49], all other studies had no participant attrition. Two studies did not report whether patients had previous orthodontic treatment [44, 49]. In addition to reporting adequate oral hygiene at baseline, two studies implemented standardized oral hygiene instruction at treatment initiation to further ensure consistency of oral conditions during treatment [35, 46]. All studies focused on individual teeth rather than patients as the unit of analysis. All RCTs employed parallel or split-mouth study designs, with the majority featuring two arms, where the control group received either no PBM treatment or sham treatment, and the experimental group received PBM treatment. Two studies with a three-arm design [39, 42] evaluated two different PBM treatment modalities in the experimental groups, comparing the effects of various PBM approaches. One study utilized a four-arm design [47], comparing the effects of PBM treatment on OIIRR in both conventional bracket and self-ligating bracket systems. Seven studies employed a split-mouth design [35, 39–41, 43, 48, 50], which ensured good baseline consistency between the groups.

PBM interventions

Compared to the orthodontic tooth movement methods, the PBM parameters and treatment protocols employed in these studies exhibited greater diversity. Detailed information regarding PBM interventions is summarized in Table 2. All studies utilized commercially available devices from various companies across different regions. Most studies reported that the irradiation type of PBM was a gallium-aluminum-arsenide (GaAlAs) diode laser, an indium-gallium-arsenide (InGaAs) diode laser, or just a diode laser, while a few studies indicated the use of light-emitting diode (LED) as the light source [41, 42, 49, 50]. Regarding the mode of laser delivery, almost all studies employed a continuous mode, with one study

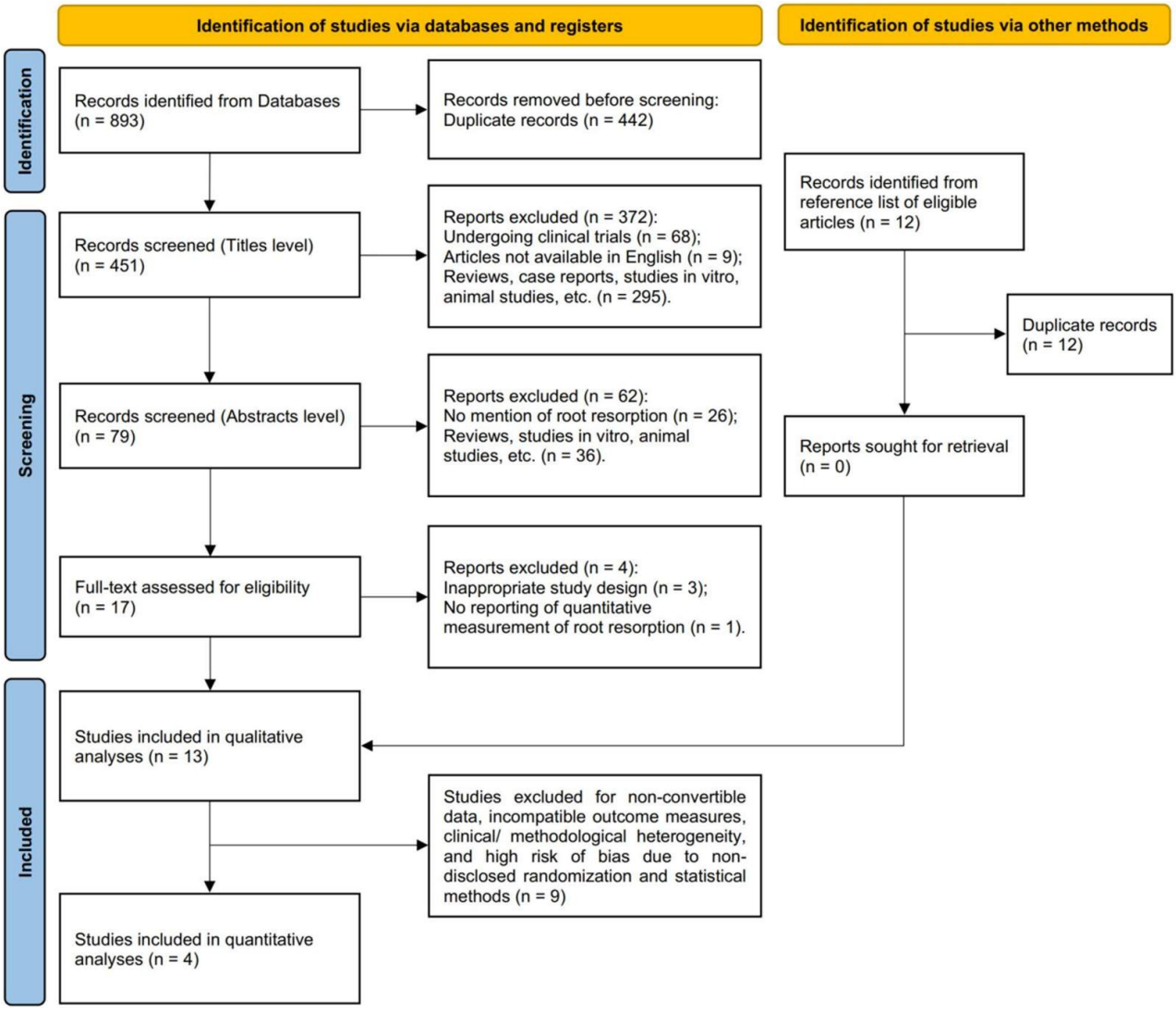


Fig. 1 PRISMA 2020 flowchart for study selection

additionally comparing pulse mode to continuous mode [39]. Three studies did not specify the mode of delivery [40, 42, 50]. The wavelengths used in these studies ranged from 660 nm to 980 nm, which can be broadly categorized into three groups: shorter wavelengths (660–670 nm) [35, 46], medium wavelengths (808–850 nm) [39, 41, 42, 44, 49, 50], and longer wavelengths (940–980 nm) [40, 43, 45, 47, 48]. Only three studies provided complete numerical values for output power, power density (irradiance), energy density (fluence), irradiation area, and irradiation time [35, 39, 46]. For some studies, missing parameters could be derived from a subset of these five parameters, but in a few cases, insufficient information precluded such derivations [41, 42, 49]. Based on the available data, the output power of PBM devices ranged from 50 mW to 2.5 W, irradiance from 20 mW/cm^2 to 4.78 W/cm^2, fluence from 0.065 J/cm^2 to 25.7 J/cm^2, and the per-session dose per tooth from 7.6 J to 108 J, indicating substantial variability across studies.

The irradiation points of these devices were typically set at multiple locations on both buccal and palatal sides. For studies using portable or large-probe devices, only one or two irradiations per dental arch might be applied [41, 43, 48]. Most studies reported the duration of light exposure per irradiation point per treatment session, while some studies reported the total irradiation time per treatment session [41, 42, 49, 50]. As for the frequency and duration of PBM treatment, the studies varied significantly: treatment frequencies ranged from once daily to monthly, with total sessions varying from 4 to 180, and treatment durations spanning from one to six months. Several studies did not specify the treatment duration due to its dependence on the overall orthodontic treatment



Fig. 2 Risk of bias of individual studies based on Cochrane Risk of Bias tool 2.0

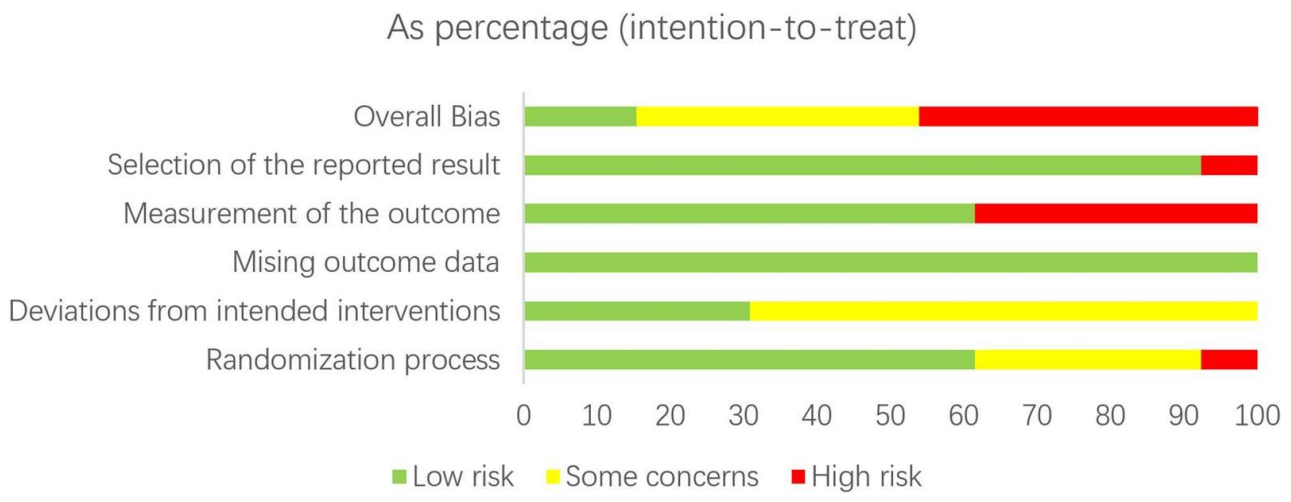


Fig. 3 Cochrane risk of bias summary

timeline [43–45, 47]. The PBM treatment duration corresponds to the follow-up period of the study.

Orthodontic interventions

All the included studies utilized fixed appliances to mediate orthodontic tooth movement, with some incorporating temporary anchorage devices (TADs) [44–46]. The orthodontic intervention methods can be categorized as follows: buccal tipping of premolars [35, 39–42], retraction of canines [43, 48] or anterior teeth [45, 50], intrusion of anterior teeth [44] or molars [46], and leveling and alignment of dentition [47, 49]. Five studies investigated buccal tipping movement of premolars achieved by a posterior segmental archwire delivering 150 g force for 28 days [35, 39–42]. One study also included an additional assessment after 6-week retention [35]. The two studies investigating upper canine retraction employed nickel-titanium closed coil springs with 150 g of force using reciprocal anchorage from anterior-posterior teeth for 3 months [43, 48]. Among the two studies applying anterior tooth retraction using power arms on different positions of wires, one utilized mini-screws to deliver 200 g of force until complete extraction space closure [45], another adopted 150 g of force for 4 months without specifying the anchorage source [50]. Only one study designed intrusion of anterior teeth, in which

Table 1 Characteristics of the included studies

Author (year)	Study design	Participants (number, mean age, gender)	Orthodontic interventions	Comparisons	Outcome measurements
Abellán et al. 2021 [46]	RCT; 2-arm parallel	20; 44.9 ± 9.6 y; 8 M/12 F	Fixed appliances; TADs (buccal and palatal) with plastic chains for 75 g of force; Maxillary first molar intrusion	PBM group (n = 10) Control group (n = 10)	RR was recorded as root volume changes by 3D rebuilding of CBCT scans
Al Okla et al. 2018 [49]	RCT; 2-arm parallel	23; 20.08 ± 1.65 y for E, 21.03 ± 2.05 y for C; N	Fixed appliances; MBT brackets; Comprehensive orthodontic treatment without tooth extraction and bracket reposition	PBM group (n = 12) Placebo group (n = 11)	RR was recorded as tooth length changes on periapical digital radiographs
Ang Khaw et al. 2018 [35]	RCT; 2-arm split-mouth	20; 15 y 9 m; 8 M/12 F	Fixed appliances; Buccal cantilever spring for 150 g/side of force; Maxillary first premolar buccal tipping; 6-week retention phase after tooth movement	PBM group (n = 20) Placebo group (n = 20) (Random side of dental arch)	RR was recorded as resorption crater volume on micro-CT images
Dehis et al. 2024 [45]	RCT; 2-arm parallel	32; 22 ± 4 y; F	Fixed appliances; NiTi closed coil springs from buccal TADs to power arms on wire for 200 g/side of force; Maxillary anterior tooth retraction	PBM group (n = 16) Control group (n = 16)	RR was graded according to the Levander and Malmgren index on CBCT images
Eid et al. 2022 [43]	RCT; 2-arm split-mouth	20; age range 15–20 y; F	Fixed appliances; NiTi closed coil springs from canines to posterior teeth for 150 g/side of force; Maxillary canine retraction	PBM group A (n = 10): PBM on days 0, 3, 7, 14, and every 2 weeks thereafter PBM group B (n = 10): PBM every 3 weeks Control group (n = 20) (Random side of dental arch)	RR was graded according to the Levander and Malmgren index on CBCT images
Goymen et al. 2020 [42]	RCT; 3-arm parallel	30; 16.27 ± 0.87 y; 14 M/16 F	Fixed appliances; Buccal cantilever spring for 150 g of force; Maxillary right first premolar buccal tipping	PBM group A (n = 10): GaAlAs laser PBM group B (n = 10): LED device Placebo group (n = 10)	RR was recorded as resorption crater volume on micro-CT images
Manoghna Subramanya et al. 2024 [50]	RCT; 2-arm split-mouth	15; age range 14–30 y; N	Fixed appliances; Power arms on wire for 150 g/side of force; Maxillary and mandibular anterior tooth retraction	PBM group (n = 15): single arch Control group (n = 15): single arch (Random quadrant of dental arch)	RR was recorded as tooth length changes on CBCT images
Nasser et al. 2023 [44]	RCT; 2-arm parallel	30; 22.4 ± 3.37 y; 13 M/17 F	Fixed appliances; Two NiTi closed coil springs from labial TADs to anterior segmental arch for 40 g/side of force; Maxillary central and lateral incisor intrusion	PBM group (n = 15) Control group (n = 15)	RR was recorded as root volume changes by 3D rebuilding of CBCT scans
Nayyer et al. 2021 [40]	RCT; 2-arm split-mouth	22; 19.1 y for M, 20.9 y for F; 11 M/11 F	Fixed appliances; Buccal cantilever spring for 150 g/side of force; Maxillary first premolar buccal tipping	PBM group (n = 22) Control group (n = 22) (Random side of dental arch)	RR was recorded as resorption crater volume on micro-CT images
Ng et al. 2018 [39]	RCT; 3-arm split-mouth	20; 16.4 ± 1.3 y for M, 16.7 ± 1.1 y for F; 10 M/10 F	Fixed appliances; Buccal cantilever spring for 150 g/side of force; Maxillary first premolar buccal tipping	PBM group A (n = 10): continuous mode PBM group B (n = 10): pulsed mode Placebo group (n = 20)	RR was recorded as resorption crater volume on micro-CT images
Saifeldin et al. 2021 [48]	RCT; 2-arm split-mouth	10; age range 18–24 y; N	Fixed appliances; NiTi closed coil springs from canines to posterior teeth for 150 g/side of force; Maxillary canine retraction	PBM group (n = 10) Placebo group (n = 10) (Random side of dental arch)	RR was recorded as tooth length changes on CBCT images
Sambevski et al. 2022 [41]	RCT; 2-arm split-mouth	20; 16.7 ± 1.2 y for M, 15.4 ± 2.8 y for F; 3 M/17 F	Fixed appliances; Buccal cantilever spring for 150 g/side of force; Maxillary first premolar buccal tipping	PBM group (n = 20) Control group (n = 20) (Random side of dental arch)	RR was recorded as resorption crater volume on micro-CT images
Shahid et al. 2023 [47]	RCT; 4-arm parallel	32; 22.41 ± 4.18 y; N	Fixed appliances; Conventional brackets or self-ligating MBT brackets; Leveling and alignment stage without tooth extraction and bracket reposition	PBM group/self-ligating bracket (n = 8) PBM group/conventional bracket (n = 8) Control group/self-ligating bracket (n = 8) Control group/conventional bracket (n = 8)	RR was recorded as blunting or shortening of the root apex (root length changes) on CBCT images

N not reported, *y* years, *m* months, *F* female, *M* male, *E* experimental, *C* control, *RR* root resorption, *TADs* temporary anchorage devices, *NiTi* nickel-titanium, *CBCT* cone-beam computed tomography, *micro-CT* micro computed tomography

Table 2 Characteristics of PBM parameters

Author (year)	Light type	Irradiation mode	Wavelength	Output power	Power density/Intensity	Energy density/Fluence	Irradiation point	Irradiation area	Irradiation time	Frequency	Per-session dose per tooth	Treatment duration
Abellán et al. 2021 [46]	Diode laser	Continuous	670 nm	150 mW	4.78 W/cm^2	11.3 J/cm^2	4 (each dental surface)	2 mm (spot diameter)	12 min (3 min/dental surface)	Days 0, 1, 2, 3, 4, 7, then monthly (12 sessions)	108 J	6 months
Al Okla et al. 2018 [49]	LED	Continuous	850 nm	N	N	0.065 J/cm^2	N	N	10 min/day (5 min per arch)	Once every day (180 sessions)	N	6 months
Ang Khaw et al. 2018 [35]	AlGaInP diode laser	Continuous	660 nm	75 mW	0.245 W/cm^2	3.6 J/cm^2	8 (4 buccal and 4 palatal)	0.26 cm^2 (spot size)	2 min (15 s/point)	Once every week (10 sessions)	7.6 J	4 weeks+ 6 weeks
Dehis et al. 2024 [45]	InGaAsP diode laser	Continuous	940 ± 10 nm	50 mW	N	7.14 J/cm^2	8 (2 labial and 6 palatal)	35 mm×8 mm (handpiece); 2.8 cm^2	Buccal: dividing the anterior segment into right and left halves, each for 60 s; Palatal: each anterior tooth for 40 s	Day of extraction, then days 7,14, 21, 28, 42, 56, then repeated biweekly till the end of retraction (more than 7 sessions)	N	First day of retraction-retraction finished
Eid et al. 2022 [43]	Diode laser	Continuous	980 nm	100 mW	N	1 J/cm^2	1 (buccal)	1 cm^2 (spot size)	8 s/point	Group A: days 0, 3, 7, 14, then every 2 weeks (9 sessions); Group B: every 3 weeks (4 sessions)	8 J	12 weeks
Goymen et al. 2020 [42]	GaAlAs diode laser	Continuous	810 nm	N	N	8 J/cm^2	10	6 mm (handpiece diameter)	N	Days 0, 3, 7, 14, 21, 28 (6 sessions)	N	4 weeks
	LED	N	850 nm	N	20 mW/cm^2	N	N	N	10 min/day	Days 0, 3, 7, 14, 21, 28 (6 sessions)	N	4 weeks
Manoghna Subramanya et al. 2024 [50]	LED	N	850 nm	N	38.49 mW/cm^2	6.9 J/cm^2	3 (all buccal)	N	6 min/day (3 min per arch)	Once every day (120 sessions)	N	4 months
Nasser et al. 2023 [44]	GaAlAs diode laser	Continuous	808 nm	250 mW	N	4 J/point	8 (4 labial and 4 palatal)	N	128 s (16 s/point)	First day of the upper incisor intrusion, then days 3, 7, 14 of the first month, then every 15 days in the second month (more than 4 sessions)	N *(8 J)*	First day of intrusion-normal overbite achieved
Nayyer et al. 2021 [40]	InGaAs diode laser	N	980 nm	100 mW	N	N	10 (5 buccal and 5 palatal)	N	10 s/point	Days 0, 3, 7, 11, 15, 28 (6 sessions)	10 J	28 days
Ng et al. 2018 [39]	Diode laser	Continuous	808 nm	0.18 W	N	1.6 J/point	8 (4 buccal and 4 palatal)	4 cm^2 (0.5 cm^2 beam area)	9 s/point	Days 0, 1, 2, 3, 7, 14, 21 (7 sessions)	N *(12.8 J)*	28 days

Table 2 (continued)

Author (year)	Light type	Irradiation mode	Wavelength	Output power	Power density/Intensity	Energy density/Fluence	Irradiation point	Irradiation area	Irradiation time	Frequency	Per-session dose per tooth	Treatment duration
	Diode laser	Pulse: 20 Hz (25 ms on/25 ms off)	808 nm	0.36 W (average 0.18 W)	N	1.6 J/point	8 (4 buccal and 4 palatal)	4 cm^2 (0.5 cm^2 beam area)	4.5 s/point	Days 0, 1, 2, 3, 7, 14, 21 (7 sessions)	N (*12.8 J*)	28 days
Saifeldin et al. 2021 [48]	InGaAs diode laser	Continuous	940 ± 10 nm	2.5 W	N	25.7 J/cm^2 (per application)	2 (1 buccal and 1 palatal)	35 mm×8 mm; 2.8 cm^2 (spot size)	30 s/point	Days 0, 1, 2, 3, 7 of maxillary canine retraction, then weekly (17 sessions)	N	3 months
Sambevski et al. 2022 [41]	LED	Continuous	850 nm	N	60 mW/cm^2	N	N	N	5 min/day	Once every day (28 sessions)	N	28 days
Shahid et al. 2023 [47]	GaAlAs diode laser	Continuous	940 nm	100 mW	N	7.51 J/cm^2	5 (all buccal)	N	6 s/tooth	Every visit until the end of the levelling and alignment stage (frequency not mentioned)	N	Whole levelling and alignment stage

N not reported, *LED* Light Emitting Diode, *AlGaInP* Aluminium Gallium Indium Phosphide, *InGaAsP* Indium Gallium Arsenide Phosphide, *GaAlAs* Gallium Aluminum Arsenide, *InGaAs* Indium Gallium Arsenide, *Italicized values* denote derived parameters not originally reported but can be calculated using Energy density multiply by Irradiation point

two mini-implants and nickel-titanium coils were used to apply a 40 g intrusive force to each side of the maxillary central and lateral incisors until a normal overbite was achieved. In another study focusing on molar intrusion, the researchers applied a 75 g intrusive force to a single elongated upper first molar using mini screws with plastic chains for a duration of 6 months. Two studies did not specify the tooth movement, instead they evaluated OIIRR after a period of comprehensive orthodontic treatment.

Characteristics of study outcomes

The included RCTs employed various methods to quantitatively assess OIIRR. In the five studies that applied buccal tipping forces to maxillary premolars [35, 39–42], the premolars were extracted after the intervention and analyzed ex vivo using micro-CT scans. The volume of craters on the root surface was calculated to represent the extent of root resorption. In other studies that extracted teeth were unavailable, OIIRR was evaluated by taking cone beam computed tomography (CBCT) or periapical digital radiographs before and after treatment. Some studies graded the severity of root resorption on CBCT images according to the Levander and Malmgren index [43, 45]. Other studies quantified root resorption by measuring length changes in root or overall tooth on CBCT images [47, 48, 50] or on periapical digital radiographs [49]. Additionally, two studies performed three-dimensional reconstruction of CBCT images to calculate changes in root volume before and after treatment [44, 46].

Calibration and reliability assessment

Five studies reported examiner training or standardized methods with repeatability metrics [42–45, 48], four of them reported high reliability based on intraclass correlation coefficient (ICC) values, whereas one missed the ICC [48]. Four trials relied on a single blinded examiner to reduce measurement error [35, 39–41], two of them quantified intra-operator error using repeated measurements and verified reliability with coefficient of variation (CV%) [35, 41], one assessed repeatability but did not provide numeric values [39], and one did not report repeatability metrics [40]. Four trials provided no information on examiner training, repeatability, or ICC [46, 47, 49, 50].

Effectiveness of PBM on OIIRR

Since the type of tooth movement and the magnitude of orthodontic force may influence root resorption [51, 52], this article categorizes the findings based on different tooth movement modalities to evaluate the effects of PBM and its parameters on OIIRR. For studies designed

with more than two arms [39, 42, 47], report only the results pertinent to the PBM treatment modality.

Among the five studies that utilized identical orthodontic devices and applied a 150 g force to induce buccal tipping of the maxillary first premolars, two studies concluded that PBM effectively reduced the volume of root resorption craters [39, 40], while the other three studies found that PBM did not mitigate OIIRR [35, 41, 42]. These studies not only discussed the overall crater volume of root resorption but also separately analyzed the resorption volume on the buccal, palatal, mesial, and distal surfaces of the roots. Nayyer et al. irradiated the premolars of the experimental group using an InGaAs diode laser (980 nm, 100 mW) on days 0, 3, 7, 11, 15, and 28 after force application, with a 10 J/tooth per-session dose and a 60 J/tooth (6 sessions) cumulative dose [40]. Compared to the control group, the PBM-treated group exhibited a significant reduction in both the mean total crater volume and the crater volumes on four axial root surfaces. Ng et al. adopted two modes of diode laser (808 nm, 0.18 W for continuous mode, 0.36 W for pulsed mode) to irradiate the premolars on days 0, 1, 2, 3, 7, 14, and 21 after force application, with a calculated per-session dose of 12.8 J/tooth and a cumulative dose of 89.6 J/tooth (7 sessions) [39]. The study confirmed that PBM treatment reduced the total volume of root resorption craters, but no significant differences were observed in the crater volumes on individual root surfaces between the PBM and control groups. This study also compared the efficacy of continuous and pulsed laser modes, finding that under the same energy density (3.2 J/cm^2), the pulsed laser group showed slightly less root resorption than the continuous laser group, although without statistical significance. Sambevski et al. performed self-administered PBM treatment using a portable LED device (850 nm, continuous mode, 60 mW/cm^2) once daily for 28 days with incalculable per-session dose and total dose [41], while no significant effect on root resorption was observed. An AlGaInP laser (660 nm, continuous mode, 75 mW) was used to irradiate the premolars once weekly for 6 weeks in Ang Khaw's study [35], with an energy density of 3.6 J/cm^2, a per-session dose of 7.6 J/tooth, and a cumulative dose of 76 J/tooth (10 sessions). The study noted a slight tendency for greater resorption crater volume in the sham group compared to the laser group without statistical significance. Similarly, premolars were irradiated using either a GaAlAs diode laser (810 nm, 8 J/cm^2) or an LED device (850 nm, 20 mW/cm^2) on days 0, 3, 7, 14, 21, and 28 after force application in Goymen's study [42]. No significant differences were found between the PBM and control groups in terms of regional and total crater volumes, and the type of light did not significantly influence root resorption outcomes.

The two studies investigating canine retraction using diode lasers with different parameters for irradiation both concluded that PBM had no significant effect on OIIRR. The first study employed a diode laser (980 nm, continuous mode, 100 mW) to treat the canines on days 0, 3, 7, and 14 after the initiation of retraction, followed by biweekly treatments, with an energy density of 1 J/cm^2 and a per-session dose of 8 J/tooth [43]. This study also tested an alternative treatment frequency, irradiating the canines every three weeks starting from the beginning of retraction, with both protocols lasting for 12 weeks. The cumulative dose of these two frequency modes was calculated as 72 J/tooth (9 sessions) and 32 J/tooth (4 sessions), respectively. The second study used an InGaAs diode laser (940 nm, continuous mode, 2.5 W) to irradiate dentition, with an energy density of 25.7 J/cm^2 per application [48]. Treatments were administered on days 0, 1, 2, 3, and 7 after the initiation of retraction, followed by weekly treatments for 3 months, totaling 17 sessions. The two studies investigating anterior tooth retraction obtained some effective results. One study applying 200 g of force utilized an InGaAsP diode laser (940 nm, continuous mode, 50 mW) with an energy density of 7.14 J/cm^2 [45]. The irradiation was performed on the day of extraction, followed by days 7, 14, 21, 28, 42, and 56, and then biweekly until the completion of retraction. The study reported a significant reduction of root resorption in the lateral incisors and the left canine. Another study applying 150 g of force used an LED device (850 nm, 38.49 mW/cm^2) with an energy density of 6.9 J/cm^2 [50]. The irradiation was performed every day for 4 months (120 sessions), and the relief of root resorption was found in the maxillary incisors.

Nasser et al. investigated the effect of PBM treatment on OIIRR during the intrusion of maxillary incisors [44]. A GaAlAs diode laser (808 nm, continuous mode, 250 mW) was used to irradiate the teeth on days 1, 3, 7, and 14 after the initiation of intrusion, followed by treatments every 15 days during the second month until the completion of intrusion. The PBM device had an energy density of 4 J/point, and the calculated per-session dose averaged 8 J/tooth (32 J for upper four incisors). The study found that PBM treatment did not have a beneficial effect on root resorption. Abellán et al. explored the role of PBM in OIIRR during molar intrusion, using a diode laser (670 nm, continuous mode, 150 mW) to irradiate the teeth on days 0, 1, 2, 3, 4, and 7 after the initiation of intrusion, followed by monthly treatments for 6 months [46]. The PBM device had an energy density of 11.3 J/cm^2, with a per-session dose of 108 J/tooth for 12 sessions. Due to extended intervals (monthly) in the later phase, the true dose accumulation was hard to calculate because biological memory effects of PBM could decay beyond the critical time window [53, 54]. The study observed a

reduction in root resorption in the PBM group compared to the control group but without a statistical significance difference.

Another two studies reported on the effects of PBM on OIIRR of anterior teeth during conventional orthodontic treatment, specifically the leveling and alignment phase [47, 49]. Shahid et al. used a GaAlAs diode laser (940 nm, continuous mode, 100 mW) with an energy density of 7.51 J/cm² to irradiate the teeth at each clinical visit [47]. Treatment continued until the completion of the leveling and alignment phase without records of exact treatment frequency. The study found that root resorption was reduced only in the left lateral incisor in the PBM group, with no significant effects observed in other teeth. Another study employed an LED device (850 nm, continuous mode) with an energy density of 0.065 J/cm², administering daily treatments until the end of orthodontic treatment, a total of 180 sessions [49]. Similarly, no clear reduction in root resorption was observed with PBM treatment.

Results of meta-analysis

Four studies that employed the same orthodontic tooth movement modality (buccal tipping of premolars) with comparable force magnitudes and outcomes were included in the meta-analysis [35, 39–41]. Although the two studies investigating canine retraction shared similar orthodontic tooth movement modalities, they were excluded due to the non-continuous rank outcome data with questionable normal distribution [43, 48]. Regarding the exclusion of two studies on anterior tooth retraction [45, 50], the first reason was heterogeneity in both orthodontic force and outcome measures, and the second reason was the high risk of bias from non-disclosed statistical methods [50].

Given the heterogeneity in PBM treatment protocols, a random-effects model was employed for all meta-analysis. Mean differences (MD) were used for effect measures because the identical outcomes among the studies were assessed using the same method, and the unit of MD was cubic millimeter (mm^3). For the only three-arm study, the pulsed and continuous PBM arms were combined into a single PBM group for the meta-analysis, as there was no statistically significant difference in efficacy between the two modes [39]. A total of 82 samples (82 maxillary first premolars) from the PBM group and 82 from the control group were extracted from the four studies. Figure 4 presented the calculated total crater volumes of the roots in the PBM and control groups. Tau² index (0.00) confirmed no significant heterogeneity among the studies, and the Cochran's Q test (p = 0.27) and I^2 index (24%) further supported the absence of heterogeneity. The mean difference in total crater volume between the PBM and control groups was – 0.13 mm^3 (95% CI: –0.17 to –0.09 mm^3, p < 0.00001), indicating that PBM treatment reduced root resorption volume. The sensitivity analysis of meta-analysis didn't change the results (p < 0.00001), suggesting a robust result (Fig. S1, S2).

To account for the varying mechanical forces exerted on different root surfaces during buccal tipping, subgroup analysis was conducted based on the buccal, palatal, mesial, and distal root surfaces. The results of the subgroup analysis are presented in Fig. 5. Tau² index (0.00) and Cochran's Q test (p > 0.05) confirmed no significant heterogeneity within each subgroup. Additionally, Cochran's Q test (p = 0.86) and I^2 index (0%) confirmed no significant heterogeneity among the four subgroups. The mean differences between the PBM and control groups for the buccal, palatal, mesial, and distal surfaces were – 0.03 mm^3 (95% CI: –0.05 to –0.01 mm^3, p = 0.003), –0.03 mm^3 (95% CI: –0.04 to –0.02 mm^3, p < 0.00001), –0.03 mm^3 (95% CI: –0.05 to –0.02 mm^3, p < 0.0001), and – 0.03 mm^3 (95% CI: –0.06 to 0.01 mm^3, p = 0.21), respectively. Except for the distal surface, the crater volumes on the other three surfaces were significantly reduced by PBM treatment. The overall pooled effect size across all subgroups was – 0.03 mm^3 (95% CI: –0.04 to –0.02 mm^3, p < 0.00001), confirming the statistical significance of the overall effect although minimal values.

The meta-analysis initially intended to conduct a subgroup analysis based on different PBM parameters. However, due to missing reports of essential parameters, there were insufficient studies within each subgroup to perform meaningful analysis. Additionally, the available

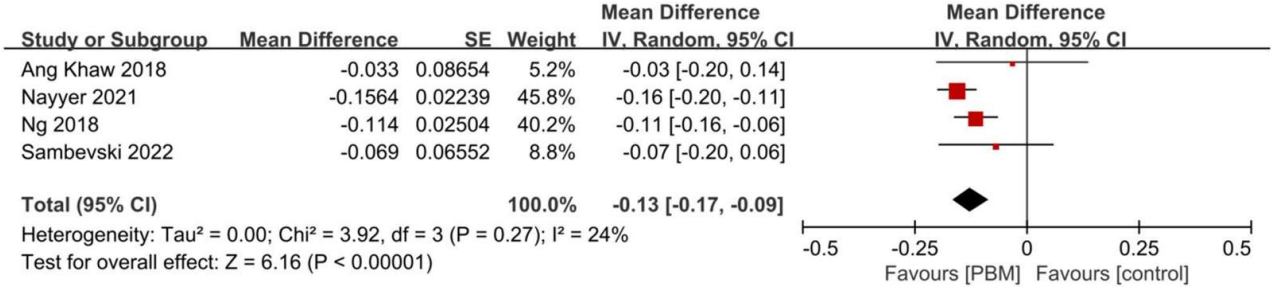


Fig. 4 Forest plot reporting mean total crater volume (mm^3) of OIIRR (within-person correlation was set as R = 0.50)

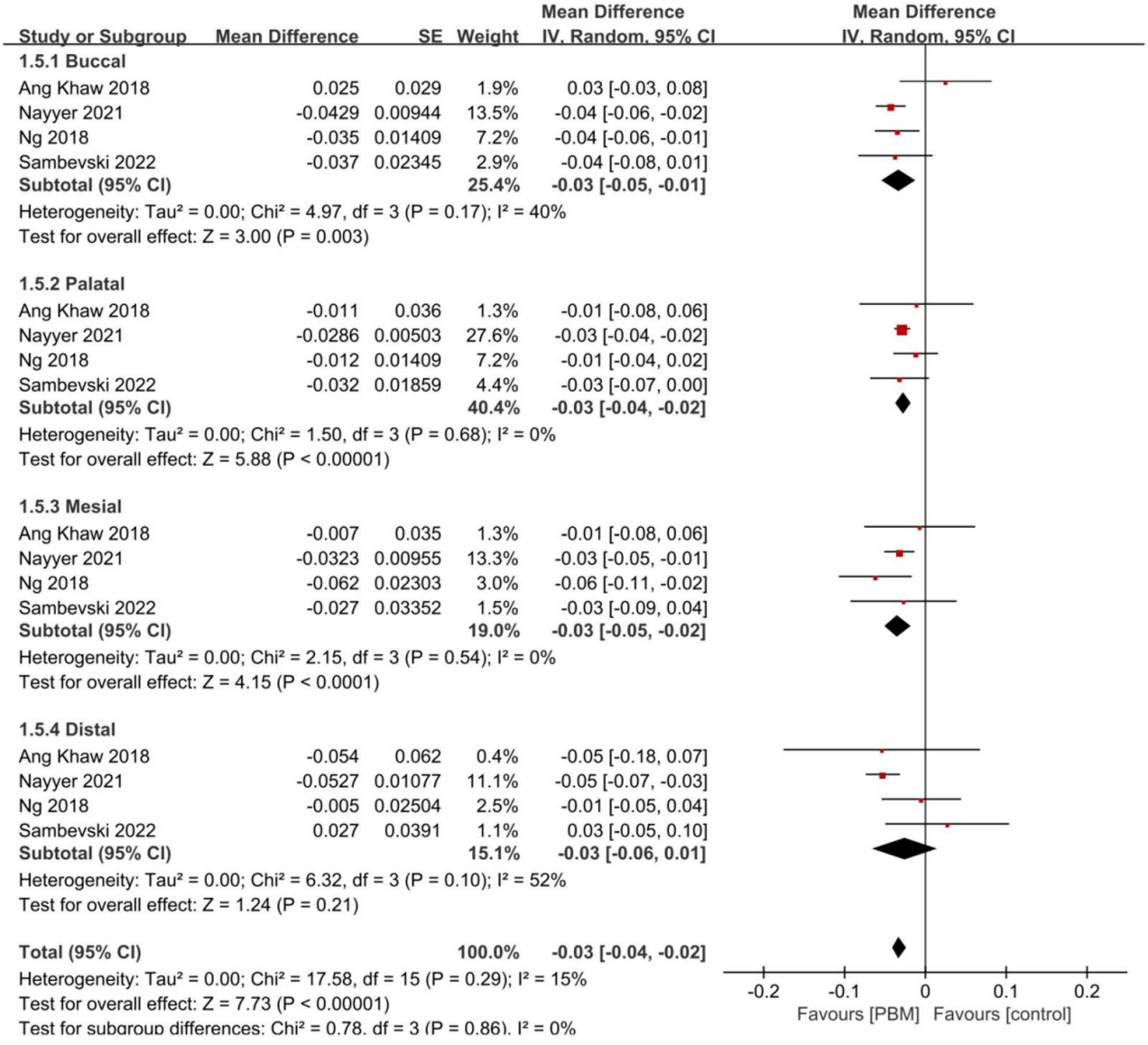


Fig. 5 Forest plot reporting mean crater volume (mm^3) of OIIRR on each axial surface of roots (within-person correlation was set as $R = 0.50$)

parameters could not be reasonably categorized to ensure an adequate number of studies within each subgroup. Consequently, subgroup analysis related to PBM parameters was not feasible. A summary of these findings was included in the qualitative analysis section.

Grading of Recommendations, Assessment, development and evaluations (GRADE)

The evidence certainty was evaluated for seven outcomes using the GRADEpro GDT online tool (Table 3), demonstrating substantial inter-rater agreement ($\kappa = 0.720$). The meta-analysis yielded low-certainty evidence, indicating limited confidence in PBM efficacy assessments. The overall evidence certainty ranged from very low to low across the studies. These findings highlight substantial limitations in the evidence base, requiring cautious interpretation of clinical implications. As detailed at the bottom of the table, evidence downgrading primarily resulted from risk of bias concerns, heterogeneity in PBM parameters and application protocols, inconsistent conclusions, and indirectness and imprecision in outcome reporting.

Discussion

As orthodontists seek reliable methods to reduce OIIRR for long, this review evaluated PBM's protective effects on OIIRR across various tooth movement types, translating existing RCT findings into evidence for clinical PBM implementations. To address the major challenge of parameter choices in PBM therapy [24, 55, 56], this review systematically summarized multidimensional PBM parameters and their corresponding therapeutic

Table 3 GRADE summary of outcome evidence certainty in meta-analysis and included records

Certainty assessment							№ of patients		Effect	Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PBM	Placebo	Absolute (95% CI)		
Total crater volumes and crater volumes on four axial root surfaces of upper first premolars in upper first premolar buccal tipping											
4	Randomised trials	Serious[a]	Serious[b]	Not serious	Not serious	None	82	82	MD for U4: **0.13 lower** (-0.17 to -0.09 mm^3)	⊕⊕◯◯ Low[a,b]	CRITICAL
Root resorption index of upper canines in canine or anterior tooth retraction											
2	Randomised trials	Serious[a]	Very serious[b,c]	Not serious	Not serious	None	26	26	NA‡	⊕⊕◯◯ Very Low[a,b,c]	IMPORTANT
Root length changes of upper canines in canine or anterior tooth retraction											
2	Randomised trials	Very serious[d]	Serious[b]	Very serious[e,f]	Very serious[f,g]	None	25	25	MD for U3: **0.02 lower** (-0.18 to 0.14 mm)	⊕◯◯◯ Very low [b,d,e,f,g]	IMPORTANT
Root volume changes of upper incisors in upper incisor intrusion											
1	Randomised trials	Serious[h]	NA†	Not serious	Not serious	None	15	15	MD for U1: **2.41 lower** (-1.06 to 5.88 mm^3) MD for U2: **2.63 lower** (-0.56 to 5.82 mm^3)	⊕⊕◯◯ Low[h]	IMPORTANT
Root length changes of upper incisors in upper incisor intrusion											
1	Randomised trials	Serious[h]	NA†	Not serious	Not serious	None	15	15	MD for U1: **0.10 lower** (-0.04 to 0.24 mm) MD for U2: **0.11 lower** (-0.10 to 0.32 mm)	⊕⊕◯◯ Low[h]	IMPORTANT
Root volume changes of upper first molars in upper first molar intrusion											
1	Randomised trials	Very serious[i]	NA†	Serious[f]	Very serious[f,g]	None	10	10	NA‡	⊕◯◯◯ Very low[f,g,i]	IMPORTANT
Root length changes in anterior teeth during conventional orthodontic treatment											
2	Randomised trials	Very serious[d]	Very serious[b,c]	Not serious	Serious[g]	None	28	27	MD⁋ for anterior teeth: **0.23 lower** (-0.38 to -0.09 mm; Shahid et al.) MD for U1-length: **0.26 higher** (-0.25 to 0.77 mm; Al Okla et al.)	⊕◯◯◯ Very low[b,c,d,g]	IMPORTANT

CI: confidence interval; **MD:** mean difference; **NA:** not applicable; **U1:** upper central incisors; **U2:** upper lateral incisors; **U3:** upper canines; **U4:** upper first premolars

a. The evidence was downgraded by one level because of serious concern about the risk of bias: more than one included record was evaluated as "some concerns".
b. The evidence was downgraded by one level because of heterogeneity in PBM treatment methodology: several PBM parameters were adopted in the included records.
c. The evidence was downgraded by one level because of inconsistent conclusions drawn from included records: the effect of PBM on OIIRR was unclear.
d. The evidence was downgraded by two level because of serious concern about the risk of bias: the two records were evaluated as "high risk".
e. The evidence was downgraded by one level because non-parametric tests do not report rank or median differences.
f. The evidence was downgraded by one level because the study didn't report the statistical analysis methods and effect size.
g. The evidence was downgraded by one level because the study didn't report confidence intervals.
h. The evidence was downgraded by one level because of serious concern about the risk of bias: the record was evaluated as "some concerns".
i. The evidence was downgraded by two level because of serious concern about the risk of bias: the record was evaluated as "high risk".
†. Not applicable because of single study.
‡. Not applicable because the data was analyzed as non-continuous rank data with non-parametric test, and its normal distribution was questionable.
‡. Not applicable because the study only provided a histogram without specific values of root resorption.
⁋. Assum the data followed a normal distribution (or lognormal distribution), and the mean and standard deviation were estimated using the median and IQR.

efficacy against OIIRR, offering novel preliminary recommendations for parameter selection in this field. To avoid interference with the results of OIIRR, the exclusion criteria for participants were built based on risk factors proven in previous studies to have potential effects on root resorption [57–63]. To ensure that radiographic evidence of OIIRR could be detected, we specified a minimum follow-up duration of 28 days (4 weeks) [6]. However, this may introduce inconsistency by allowing inclusion of studies focusing on either short-term or long-term effects of root resorption.

The meta-analysis of the four studies investigating premolar buccal tipping revealed that the PBM group demonstrated statistically significant reductions in root resorption crater volume compared to the controls, both for the entire root and across all four axial root surfaces. These four studies lasted 4 weeks, and early OIIRR could be detected because of the usage of high-resolution micro-CT for measurement [64, 65]. Nevertheless, these studies provide information only on short-term effects. The estimated mean difference in total root resorption volume was – 0.13 mm^3 (95% CI: –0.17 to –0.09 mm^3). Although the difference is not negligible relative to an average total resorption volume of 0.5 mm^3, the absolute reduction is still small, and its clinical significance should be interpreted with caution. As noted earlier, because the follow-up was only 4 weeks, our quantitative analysis focuses on early and subtle OIIRR, its long-term effects need to be confirmed in future studies. The protective effect of PBM showed no significant variation among the four axial surfaces of the roots, suggesting that the therapeutic efficacy may be independent of location-specific stress distribution patterns. Since diverse orthodontic interventions may influence root resorption outcomes [6, 52, 66], the results should be discussed by category rather than pooling. Previous research has identified intrusive tooth movement as potentially the strongest predictor for increased incidence and severity of OIIRR [67, 68]. However, only two RCTs in the review specifically investigated the intrusive movement of incisors or molars [44, 46], suggesting either insufficient attention to this high-risk tooth movement type or challenges in experimental design. Less precise and inconsistent assessment of OIIRR in canine retraction also leads to its inability to be included in quantitative analysis [43, 45, 48].

This review also revealed distinct parameter-efficacy relationships across different tooth movement types of PBM therapy. The possible suitable parameter ranges and

Table 4 Possible suitable parameter ranges of PBM therapy for OIIRR and the supporting data

Parameters	Values from positive results	Values from negative results	Possible suitable ranges
Output power	180 mW [39]; 100 mW [40]; 50 mW [45]; 100 mW [47]	75 mW [35]; 100 mW [43]; 2500 mW [48]; 250 mW [44]; 150 mW [46]	100–200 mW
Energy density	7.14 J/cm^2 [45]; 7.51 J/cm^2 [47]; 6.9 J/cm^2 [50]	3.6 J/cm^2 [35]; 1 J/cm^2 [43]; 11.3 J/cm^2 [46]; 0.065 J/cm^2 [49]	5–10 J/cm^2
Per-session dose per tooth	12.8 J/tooth [39]; 10 J/tooth [40]	7.6 J/tooth [35]; 8 J/tooth [43]; 8 J/tooth [44]; 108 J/tooth [46]	10–100 J/tooth
Wavelength	808 nm [39]; 980 nm [40]; 940 nm [45]; 940 nm [47]; 850 nm [50]	660 nm [35]; 850 nm [41]; 810/850 nm [42]; 980 nm [43]; 808 nm [44]; 670 nm [46]; 940 nm [48]; 850 nm [49]	Not applicable

Values from positive/negative results: parameters from the experiments that did/did not demonstrate the PBM efficacy on OIIRR.

the supporting basis are listed in Table 4 for readers' reference. For premolar buccal tipping, there is no clear difference in the wavelengths of the PBM devices, except for one study that used the less common shorter wavelength (660 nm) [35]. In contrast, the output power and per-session dose per tooth appear to play more significant roles in treatment efficacy. Besides, a tapered frequency regimen may play an important role. The protocol represents shorter irradiation intervals (probably no longer than one week) during the initial treatment phase and followed by progressively extended intervals (probably no longer than two weeks) in later stages [69]. No clear relationship could be summarized between cumulative doses and PBM efficacy due to similar dose without a trend among the studies (Nayyer et al. 60 J/tooth, Ng et al. 89.6 J/tooth, Ang Khaw et al. 76 J/tooth). Based on these findings, for premolar buccal-tipping movements, wavelength appears to have unclear influence on PBM efficacy, whereas higher output power (≥ 100 mW) and greater per-session dose (≥ 10 J/tooth) may demonstrate more favorable therapeutic outcomes. This effect may be further potentiated by a tapered irradiation protocol of frequency.

For canine and anterior tooth retraction, studies present a more complex pattern. Despite sharing similar longer wavelengths and treatment duration, the two negative studies exhibited 25-fold variations in output power (100 mW vs. 2.5 W) and energy density (1 J/cm² vs. 25.7 J/cm²) [43, 48]. Besides, the study using the lower value administered a per-session dose lower than 10 J/tooth that mentioned above [43]. The negative outcomes probably come from their parameters falling outside a possible optimal range, one being too high and the other too low. Moreover, current guidelines typically stipulate that the output power should not exceed 500 mW in general, as higher intensities may no longer qualify as "low-level" and could pose a greater risk of thermal damage [16, 70]. In contrast, the two positive studies on anterior tooth retraction employed a moderate energy density (7.14 J/cm², 6.9 J/cm²) that may fall within the therapeutic window [45, 50]. The study with the lowest output power (50 mW) maintained a tapered irradiation protocol of frequency, suggesting its possible beneficial role [45]. Although neither reported exact dose values, both protocols imply the potential existence of effective cumulative PBM exposure due to the possible long treatment period. Overall, the wavelength shows unclear effects on treating canine OIIRR. A tapered irradiation protocol from weekly to biweekly may be preferable, which also aligns with the time-dependency effect of PBM therapy [53]. Appropriate therapeutic ranges likely exist for both output power and energy density, within which the cumulative dose shouldn't be ignored.

For incisor and molar intrusion, both studies failed to demonstrate significant PBM efficacy [44, 46]. Given the anatomical differences between incisor and molar roots and variations in intrusive forces, a direct comparison of PBM parameters proved challenging. Considering the output power exceeding 100 mW in both studies, what causes negative results may be inadequate per-session dose or higher output power as 250 mW [44]. Per-session dose of 8 J/tooth is not in line with the thresholds of 10 J/tooth [44], and 108 J/tooth may suggest that excessively high single doses potentially diminish therapeutic effects [46]. Besides, monthly intervals in the later phase may nullify the proposed advantages of tapered protocol.

Among conventional orthodontic treatment [47, 49], the wavelength parameters showed no distinctive features. The study reporting positive effects on left lateral incisors employed an output power (100 mW) and energy density (7.51 J/cm²) that might fall within the previously suggested possible ranges, though the exact per-session dose remained unspecified [47]. Note that the statistical significance in the study was achieved only at a single tooth position, potentially reflecting a measurement error. The other study utilized an extremely low energy density (0.065 J/cm²), and the cumulative dose was not appreciable despite daily treatment, both of which likely accounted for its negative findings regarding PBM efficacy [49].

The included RCTs exhibited considerable diversity in PBM parameters, except for the sole consistency of continuous mode, which has been previously validated as effective in orthodontic tooth movement and root

resorption [22, 71, 72]. Additionally, the potential differential effects of diode laser versus LED light sources remain unclear [18, 73]. Although all LED-based studies in our analysis reported negative outcomes, the variations in intervention methods precluded meaningful clinical interpretation. It is important to note that the units or applicability of the parameters, whether directly reported or derived, were not entirely consistent. For example, some studies calculated power density based on a single irradiation point rather than per square centimeter [40, 44]. Overall, compiling a standardized set of comparable PBM parameters proved challenging.

Several factors may limit the generalizability of our findings. The meta-analysis can only be applied to the buccal-tipping movement of premolars due to a large ineligibility of studies, while the clinical relevance of this specific movement needs further investigation, and the minimal absolute magnitude of the differences warrants further evaluation for clinical significance. The different quality of calibration and reliability with the pooled studies may introduce measurement error, and the results should be interpreted with caution. The limited number of studies for each tooth movement type precluded subgroup analysis based on PBM parameters, leaving the crucial problem of optimal PBM parameters for OIIRR incompletely solved. The narrative synthesis of PBM parameters according to different orthodontic movements was still limited by heterogeneities in parameter selection and incomplete reports across studies. Participant characteristics like age, sex, and ethnicity were variably reported and underrepresented, constraining external validity. In studies of anterior tooth intrusion, anterior tooth retraction, and alignment and leveling, multiple teeth per patient included in each group may introduce within-patient clustering. Varied outcome measurement methods, small sample sizes, and different study designs may increase imprecision and the potential for unit-of-analysis issues. Several studies also had short follow-up periods, limiting assessment of longer-term resorption risk and durability of effects. Although RCTs represent high-level evidence [74, 75], the RoB assessment revealed substantial methodological concerns, making the pooled estimate vulnerable to bias and potentially overstating the true effect. Coupled with a GRADE rating of low certainty, this substantially limits our confidence in the magnitude and generalizability of the findings. Besides, the time-unrestricted literature search showed that clinical trials on PBM's effects on human OIIRR have only emerged in the last decade and remain scarce. Our restriction to English-language publications may also have introduced language bias. More high-quality, well-designed RCTs are still needed to strengthen the evidence base in this field.

Conclusions

This systematic review and meta-analysis of RCTs indicates that PBM can reduce the severity of OIIRR during buccal-tipping movement of maxillary first premolars, especially in the early stages of treatment. For other tooth movement types, such as canine retraction, incisor or molar intrusion, the limited and heterogeneous evidence precludes consistent conclusions about efficacy. Exploratory findings suggest that PBM effects may depend on device output power, energy density, and per-session dose per tooth. The suggested parameter ranges should be considered preliminary rather than definitive thresholds. Wavelength appears to have a minimal impact on treatment efficacy. Cumulative dose and tapered irradiation frequency may influence outcomes but require confirmation. The parameters can sometimes be mutually calculated and can be considered comprehensively when the report is complete. Future studies can specifically validate the cumulative-dose-based PBM effects and develop parameter optimization strategies for different tooth positions and orthodontic forces.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12903-025-07512-y.

Supplementary Material 1.

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Supplementary data

Detailed search strategy of PubMed, Web of Science, Cochrane Library, Embase via Ovid, MEDLINE via Ovid, and Scopus. Figures of sensitivity analysis of meta-analysis.

Authors' contributions

G.Z.: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing - Original Draft, Visualization. Z.L.: Methodology, Validation, Formal analysis, Visualization. C.M.: Supervision, Writing - Review & Editing. Y.Y.: Resources, Supervision. Z.S.: Conceptualization, Methodology, Validation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]Division of Paediatric Dentistry and Orthodontics, Faculty of Dentistry, The University of Hong Kong, Hong Kong SAR, China
[2]Division of Applied Oral Sciences and Community Dental Care, Faculty of Dentistry, the University of Hong Kong, Hong Kong SAR, China

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